Technical University of Denmark

SPRING, 2025
Campus: Lyngby

MODEL-BASED MACHINE LEARNING

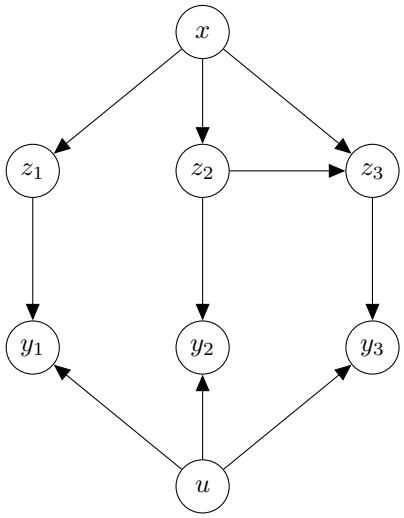
Mini-test 1

(Time allowed: 60 MINUTES)

Name: _____
Student ID: _____

Part 1

Consider the following probabilistic graphical model (PGM):



1. (10%) Write the factorization of the joint distribution according to the PGM described above.

$$p(x)p(u)p(z_1|x)p(z_2|x)p(z_3|z_2,x)p(y_1|z_1,u)p(y_2|z_2,u)p(y_3|z_3,u)$$

2. (20%) Which of the following expressions correspond to valid expressions for $p(z_3|z_2,u)$ (more than option may the correct):

(a) $\frac{\sum_x p(x)p(z_2|x)p(z_3|z_2,x)}{\sum_x p(x)p(z_2|x)}$

3. (15%) Classify the following statement as “True", “False", or “Cannot prove":

(a) $u \perp\!\!\!\perp z_3 | \{y_3, x\}$ - cannot prove

(b) $x \perp\!\!\!\perp y_1 | \{z_1, y_2, y_3\}$ - cannot prove

Part 2

Consider the following generative process that models observed count data $\{y_{i,n}\}_{n=1}^{N_i}$ for each individual i using a hierarchical modelling approach:

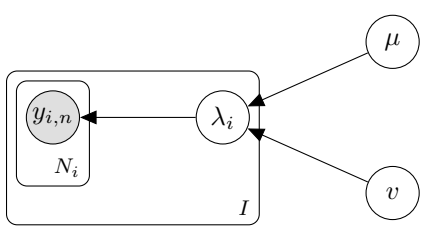
1. Draw $\mu \sim \mathcal{N}(0, 1)$
2. Draw $v \sim \text{Inverse-Gamma}(1, 1)$
3. For each individual $i \in \{1, \ldots, I\}$:
   (a) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\mu, v)$
   (b) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:
      i. Draw observed count $y_{i,n} \sim \text{Poisson}(\exp(\lambda_i))$

Make sure that you understand it well.

1. (15%) Write the factorization of the joint distribution according to the model described above.

$$p(\mu)p(v)\prod_{i=1}^{I} p(\lambda_i|\mu, v)\prod_{n=1}^{N_i} p(y_{i,n}|\lambda_i)$$

2. (15%) Draw the PGM representation corresponding to the generative process described above.



3. (10%) Considering the generative process of the model presented above, classify the following statements as True (T) or False (F).

(a) $p(\mu,\ \boldsymbol{\lambda}) = \int_v \int_{\mathbf{y}} p(\mu, v, \boldsymbol{\lambda}, \mathbf{y})$ - F

(b) $p(\boldsymbol{\lambda}|\ \mathbf{y}, \mu, v) \propto p(\mu, v, \boldsymbol{\lambda}, \mathbf{y})$ - T

(c) $p(\mu|\ \mathbf{y}, \boldsymbol{\lambda}, v) \propto p(\mu, v, \boldsymbol{\lambda}, \mathbf{y})$ - T

(d) $p(\mu|\ \lambda) = \frac{\int_v p(\mu)p(v)p(\boldsymbol{\lambda}|\mu,v)}{\int_v \int_{\boldsymbol{\lambda}} p(\mu)p(v)p(\boldsymbol{\lambda}|\mu,v)}$ - F

4. (15%) Suppose now that you also have access to covariates $\mathbf{x}_i$ that describe characteristics of the each individual i that could help explain his/her latent log-rate λ_i. You may assume that the relationship between the mean of λ_i and $\mathbf{x}_i$ is linear in parameters $\boldsymbol{\beta} = \{\beta_1, \beta_2, \ldots, \beta_D\}$. Which of the following generative processes corresponds to this enhanced model?

Option X:

(a) Draw $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma\mathbf{I})$

(b) Draw $v \sim$ Inverse-Gamma$(1, 1)$

(c) For each individual $i \in \{1, \ldots, I\}$:

  (i) Draw individual-specific (log-)rate $\lambda_i \sim \mathcal{N}(\boldsymbol{\beta}^T \mathbf{x}_i, v)$

  (ii) For each observation $n \in \{1, \ldots, N_i\}$ of the individual i:

    A. Draw observed count $y_{i,n} \sim$ Poisson$(\exp(\lambda_i))$